

Lecture #	Date	Day	Assignment	Papers to Read and Write a Report On	Quiz	Category	Topics	Reference Videos
1	5-Jan	Monday	Setting up colab and AWS and testing the system works	Towards felicitous decision making: An overview on challenges and trends of Big Data + Big Data Systems: A Software Engineering Perspective		Introduction	Overview of the class+ Understanding Big Data	https://youtu.be/0uc9z8-FmVk
2	7-Jan	Wednesday					Applications Of Big Data: Wildfire + Transit Analysis + Emergency Response	
3	12-Jan	Monday	Assignment 1: SQL	Anatomy of a Database System		Storage	Overview of databases and their anatomy+ACID Properties	https://www.youtube.com/watch?v=pomxIOFvcQs
4	14-Jan	Wednesday		Relational Databases : SQL and Joins and access			https://youtu.be/HXV3zeQKqGY	
5	21-Jan	Wednesday		Access Path Selection and Joins+ Optimistic Concurrency			Relational Databases: Indexes, Locks and Concurrency -00	https://youtu.be/ihXf8p-UwU
Classes Canceled								
6	2-Feb	Monday	Assignment 2: MongoDB	columnstoresfntdbs (chapters 1, 2 and 3)	Quiz 1	Storage	Distribution/Partitioning: Fault Tolerance+ CAP Theorem	https://www.youtube.com/watch?v=R_Fxz14tr2M
7	4-Feb	Wednesday					Column vs Row Oriented Databases	https://www.youtube.com/watch?v=Vw1fCeD06YI
8	9-Feb	Monday		Introducing NoSQL Databases: Need and Differences/ MongoDB			https://www.youtube.com/watch?v=QwevGzVu_zk	
9	11-Feb	Wednesday	Assignment 3: Influx and Pandas	Graph DataBase Paper		Storage	BigTable and DynamoDB	BigTable: https://www.youtube.com/watch?v=KaRbKdMnuc DynamoDB: https://www.youtube.com/watch?v=OfzPHXsYqNE
10	16-Feb	Monday					Graph Databases	https://www.youtube.com/watch?v=GekQgFZm7mA
11	18-Feb	Wednesday					Overview of Pandas and Parquet	Pandas: https://www.youtube.com/watch?v=vmEHCJofslg
12	23-Feb	Monday	Assignment 4: Hadoop and MapReduce	Orchestrating Big Data		Batch Processing	Project Overview and Discussions	Parquet: https://www.youtube.com/watch?v=pFS-FScophU
13	25-Feb	Wednesday					Map Reduce	https://www.youtube.com/watch?v=b-lvmXoO0bU
14	2-Mar	Monday					Map Reduce	
15	4-Mar	Wednesday		map-reduce-bookchapter.pdf + MapReduce: Simplified Data Processing on Large Clusters	Quiz 2	Batch Processing	Spark	https://www.youtube.com/watch?v=lQfG0faDrzE
Spring Break								
16	16-Mar	Monday	Assignment 4: Hadoop and MapReduce	map-reduce-bookchapter.pdf + MapReduce: Simplified Data Processing on Large Clusters		Streaming	Dask	https://www.youtube.com/watch?v=EybGGlbiLpI
17	18-Mar	Wednesday	Assignment 5: Spark	ComparisonOfApproachesToLargeScaleDataAnalysis.pdf			Overview (Dataflow Model) + Kafka and Pulsar and ZooKeeper	
18	23-Mar	Monday					Kafka and Pulsar and ZooKeeper	Kafka: https://www.youtube.com/watch?v=JaIUUBKdcA0 Pulsar: https://www.youtube.com/watch?v=IAVPIY1PRDo
19	25-Mar	Wednesday			spark.pdf	Quiz 3	SparkStreaming	https://www.youtube.com/watch?v=UuRhEmqghRM
20	30-Mar	Monday		Storm			https://www.youtube.com/watch?v=0mIEUibitzk	
21	1-Apr	Wednesday	Assignment 6: Kafka and Containers	Kafka: a Distributed Messaging System for Log Processing		Analytics and Real-world Use Cases	Apache Flink	https://www.youtube.com/watch?v=G77m6Ou_kfA
22	6-Apr	Monday					Clustering	https://www.youtube.com/watch?v=eg8DIYwdMyg
23	8-Apr	Wednesday					Borg, Omega, and Kubernetes	Pagerank
24	13-Apr	Monday	Assignment 7 Visualization	Efficient Data Management for Intelligent Urban Mobility Systems		Project Presentations	Anomaly Detection	https://www.youtube.com/watch?v=12Xq9OLDQwQ
25	15-Apr	Wednesday					Project Presentations 1	None
26	20-Apr	Monday					Project Presetation 2	